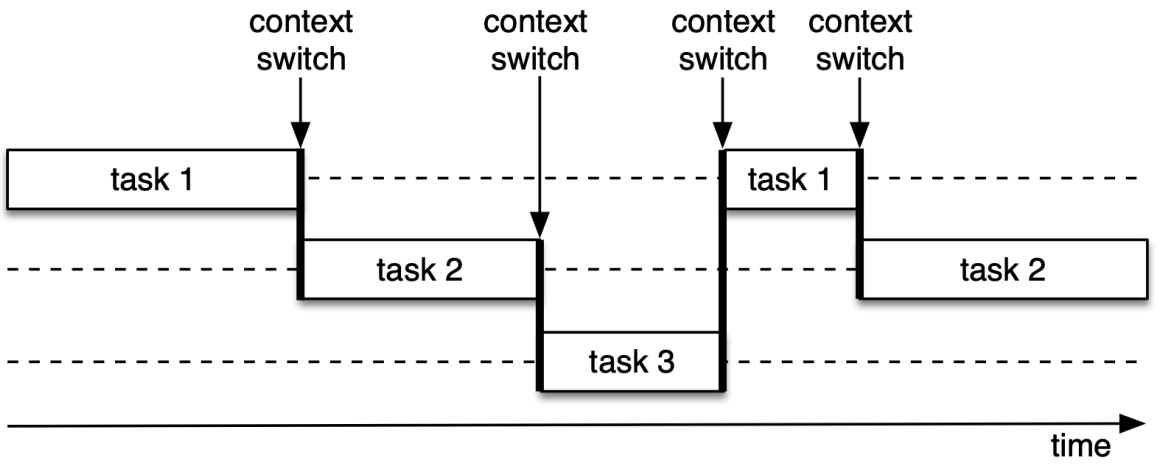


2020 Paper

Question 1

Multitasking is a technique that allows multiple tasks (you may be more familiar with the terms program, thread or process) to execute concurrently on a single processor. One processor can execute the instructions from only one task at a time, so multitasking is implemented by executing each task for a period of time before switching to another task. In this way, time executing on the processor is shared between multiple tasks, as illustrated in the diagram below.



A “context switch” from one task to another is commonly implemented by using a timer interrupt to regularly interrupt the running task. The interrupt handler for the timer will then save the state of the current task, choose the next task to execute, restore the state of the chosen task and resume its execution.

(a). Describe the state information that should be saved and restored to facilitate multitasking as outlined above, basing your answer on your knowledge of an ARM microprocessor that you are familiar with and of interrupt handling.

answer: the state information should be saved / pushed onto the system stack

Question 2

Consider an analogue baseband signal with a maximum frequency of 5MHz.
(a) Choose asuitable sampling frequency, explaining your choice, then show the frequency spectrum of the digitised signal (i.e., the Discrete-Time Fourier Transform - DTFT).
(b) Assume that after digitisation, with 12 bits per sample, you have the following initial sequence of bits: 011001010011. Show the time-domain plot of the signal when it si modulated first with a 4-ASK and then with an 8-PSK, indicating the correct duration of each time slot.
(c) Assuming the digitised signal is transmitted at a bit rate of 10 Mb/s using the 4-PSK modulation, show the frequency domain plot of the modulated signal, indicating its bandwidth, if it is modulated with a 50 MHz carrier.
(d) You now need to multiplex together four of these 10 Mb/s signals using frequency division multiplexing. The first signal is modulated with a 2-PSK, the second with a 4-PSK, the third with an 8-PSK and the fourth with a 16 - QAM. Assuming the total available bandwidth is 250 MHz, what is the maximum guard band that you can use ni
the system?

Question 3

(a)
Two E-MOSFETS are connected in series across a 10V supply. The 'upper' MOSFET has its drain connected directly to its gate. The 'lower' MOSFET is unbiased. The input to the circuit is taken to be the gate of the lower MOSFET and the output is taken to be at the drain of the lower MOSFET. Draw the circuit diagram.

(b)
The upper MOSFET has a resistance of 2 K when its gate-source voltage is 10V. The lower MOSFET has a resistance of 1500 when the input voltage is 3V. What are the outputs of the circuit when the inputs to the circuit are OV and 3V, respectively

Question 4

(a) A combinational circuit is specified by the following three Boolean functions:

$$F1(A, B, C) = \Sigma(1, 4, 6)$$
$$F2(A, B, C) = \Sigma(3, 5)$$
$$F3(A, B, C) = \Sigma(2, 4, 6, 7)$$

Implement the circuit with a decoder constructed with NAND gates and NAND or AND gates connected to the decoder outputs. Use a block diagram for the decoder. Minimise the number of inputs in the external gates.

(b) Design a combinational circuit that compares two 4-bit numbers to check if they are equal. The circuit output is equal to 1 if the two numbers are equal and 0 otherwise.
Provide a boolean algebra equation of your solution and a circuit diagram.